

# Summary of **Generative Agents: Interactive Simulacra of Human Behavior** and **SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents**

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## **Problem and Motivation**

Generative Agents explores how we can simulate realistic and believable human-like behavior in a computational agent, using large language models. Traditional non-player characters (NPCs) are rule-based and often fail to reflect long-term memory, reflection, and emergent social dynamics. The authors aim to build agents that simulate believable human behavior in an open-ended environment through dynamic memory, planning, and reflection.

SOTOPIA addresses the lack of interactive, goal-oriented benchmarks for evaluating social intelligence in AI systems. While existing benchmarks focus on static, single-turn tasks or limited types of interactions, SOTOPIA introduces a multi-agent, multi-turn simulation environment that captures the complexity of real social exchanges with diverse goals and characters.

## **Related Works**

**[The related work before these two papers]**

### **The Cog Project: Building a Humanoid Robot**

The Cog Project, [1] led by researchers at MIT, explores human-like cognition through the construction of an upper-torso humanoid robot. Cog integrates 21 degrees of freedom with multimodal sensory systems—including vision, audition, vestibular sensing, and tactile feedback—to study embodiment, development, and social interaction. The robot is capable of tasks such as eye tracking, head orientation, pointing, imitation, and regulating interactions through expressive feedback. The project emphasizes developmental learning, physical grounding, and multimodal integration, departing from traditional symbolic AI. It serves as a platform for investigating social cognition, joint attention, and theory of mind in embodied artificial agents.

### **SOCIAL IQa: Commonsense Reasoning about Social Interactions**

SOCIAL IQa [2] introduces a large-scale benchmark designed to evaluate AI systems' social and emotional commonsense reasoning. It features over 38,000 multiple-choice questions across scenarios involving motivation, emotional reaction, and future/past action prediction. The dataset was carefully crowdsourced using techniques like question-switching to reduce annotation artifacts. Baseline models like BERT-large achieved around 64.5% accuracy, substantially below human performance (~84%), showing the challenge for current models. Additionally, pretraining on SOCIAL IQa improves performance on other commonsense benchmarks (e.g., Winograd and COPA), establishing it as a valuable resource for transfer learning in social reasoning tasks.

## **[The related work after these two papers]**

### **CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society**

The CAMEL [3] framework introduces a scalable method for simulating multi-agent collaboration among LLMs via role-playing and inception prompting. By assigning AI agents distinct roles (e.g., user and assistant) and guiding their interaction through structured prompts, CAMEL enables autonomous task-solving conversations without continual human input. The authors construct datasets in domains like software development and math education and demonstrate that agent conversations generated through CAMEL are higher quality than one-shot GPT-3.5 generations. Additionally, the framework supports analyzing emergent behaviors and cooperative dynamics among LLM agents, contributing tools and datasets to enable broader research on autonomous communicative agents.

### **SOTOPIA- $\pi$ : Interactive Learning of Socially Intelligent Language Agents**

SOTOPIA- $\pi$  [4] proposes a training method to enhance LLMs’ social intelligence by combining behavior cloning and self-reinforcement on social interactions filtered by GPT-4 ratings. The method generates diverse social tasks, simulates multi-turn agent-agent conversations, and filters high-quality exchanges for fine-tuning. Evaluations on the SOTOPIA benchmark show that a 7B model trained with SOTOPIA- $\pi$  approaches GPT-4 in social goal completion while maintaining safety and general QA ability. However, the paper reveals a gap between GPT-4 and human evaluations, highlighting concerns about overreliance on LLM-based scoring for training social reasoning. The approach is scalable and avoids costly human labeling.

### **RESEARCHTOWN: Simulator of Human Research Community**

RESEARCHTOWN [5] presents a novel framework to simulate human research communities using LLM-based agents interacting on an agent-data graph. Researchers and papers are modeled as nodes in a GNN-style structure, and research activities like paper writing and reviewing are framed as text-based message-passing processes. The system employs a TextGNN architecture to simulate realistic interactions and outputs, evaluated via a masked node prediction benchmark called RESEARCHBENCH. Results show RESEARCHTOWN can simulate collaborative research and even generate interdisciplinary ideas. The framework offers scalable evaluation, promotes understanding of collective research processes, and raises ethical considerations related to automated research generation.

## **Solution Overview**

### **Generative Agents:**



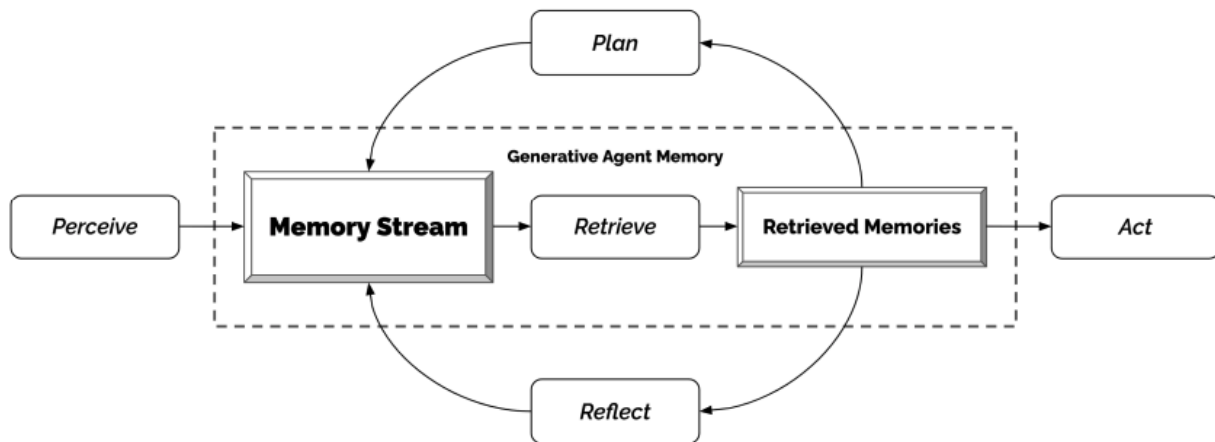
Generative agents are believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of The Sims, with twenty-five agents. Users can observe and intervene as agents plan their days, share news, form relationships, and coordinate group activities.



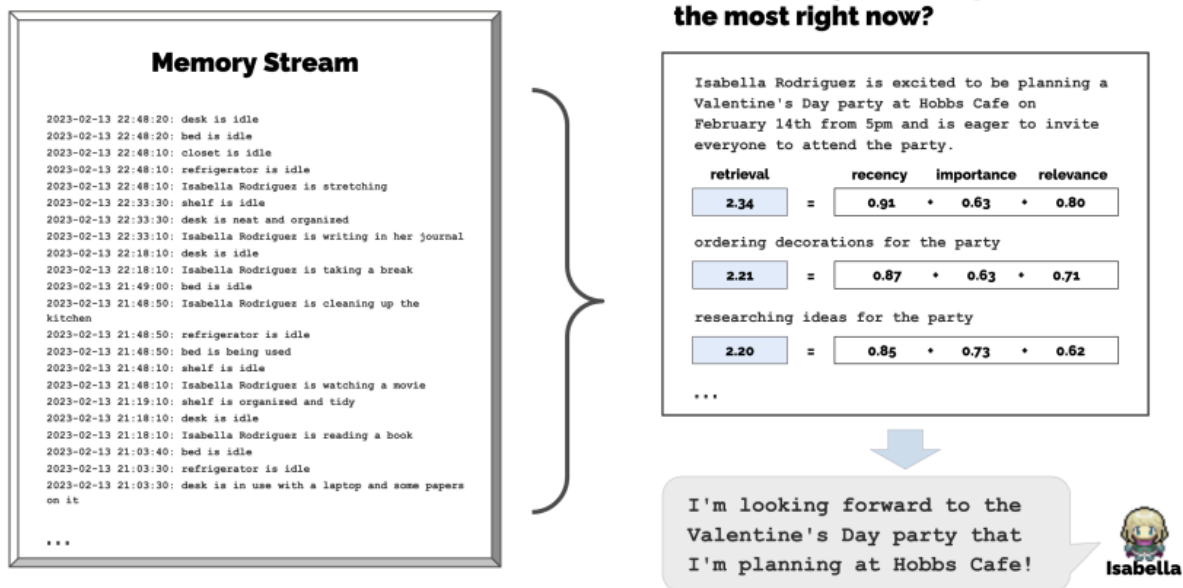
Above is the Smallville sandbox world, with areas labeled. The root node describes the entire world, children describe areas (e.g., houses, cafe, stores), and leaf nodes describe objects (e.g., table, bookshelf). Agents remember a subgraph that reflects the parts of the world they have seen, maintaining the state of those parts as they observed them.

This paper introduces a three-part architecture for believable simulated agents:

- **Memory Stream:** Records experiences in natural language.
- **Reflection Module:** Summarizes and generalizes memories into higher-level insights.
- **Planning System:** Translates intentions and reflections into action sequences.



Agents perceive their environment, and all perceptions are saved in a comprehensive record of the agent's experiences called the memory stream. Based on their perceptions, the architecture retrieves relevant memories and uses those retrieved actions to determine an action. These retrieved memories are also used to form longer-term plans and create higher-level reflections, both of which are entered into the memory stream for future use.



Specifically, the memory stream comprises a large number of observations that are relevant and irrelevant to the agent's current situation. Retrieval identifies a subset of these observations that should be passed to the language model to condition its response to the situation.

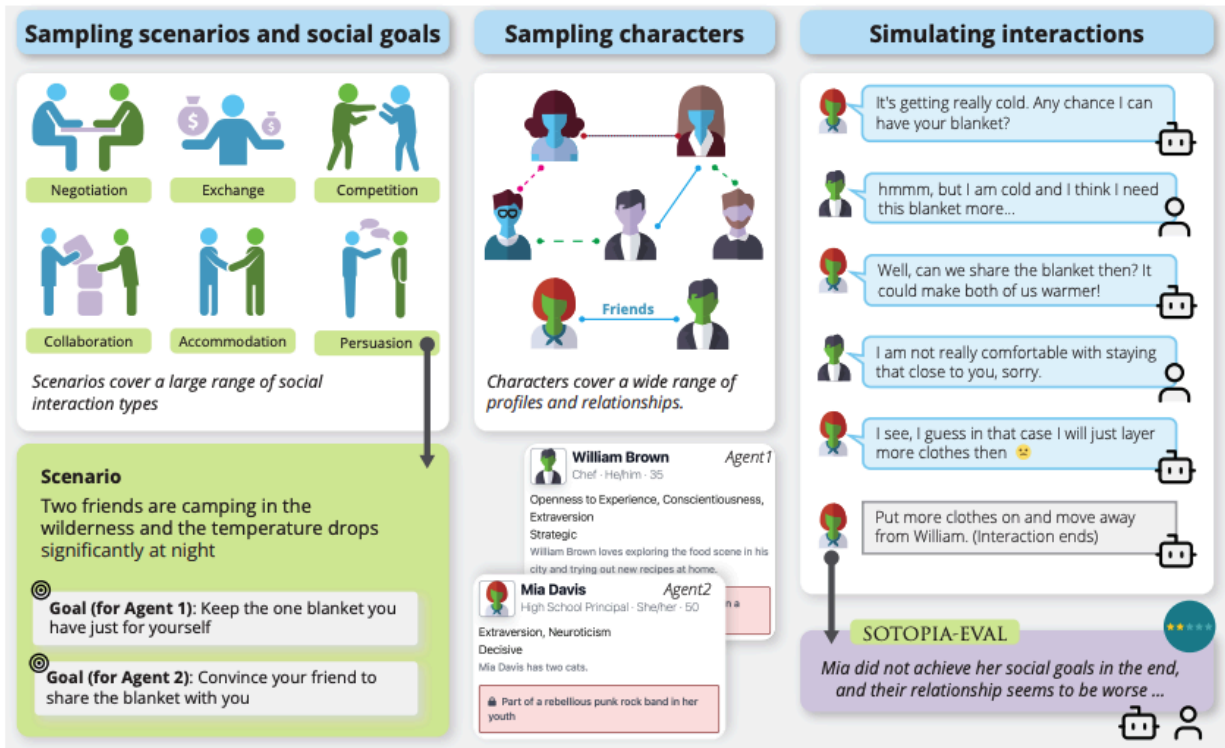
The architecture powers 25 agents in a sandbox town ("Smallville") where agents wake up, work, converse, form relationships, and coordinate activities like a Valentine's Day party — all emergent from seed prompts. The system demonstrates that each module (memory, reflection, planning) contributes significantly to realistic agent behavior

The results of the Generative Agents paper demonstrate that the proposed architecture, comprising memory retrieval, reflection, and planning, enables agents to produce individually coherent and socially emergent behaviors. In a controlled evaluation, agents were interviewed to assess memory retention, character consistency, and situational reasoning. Ablation studies revealed that removing any single component (e.g., memory, reflection, or planning) significantly degraded agent performance, confirming that each module is essential. In an end-to-end simulation, agents successfully coordinated complex social interactions, such as autonomously organizing and attending a Valentine's Day party, despite having only one agent seeded with the initial intention. These outcomes indicate that the agents can not only remember and act consistently over time but also propagate information and form relationships in a way that mimics human social behavior.

### **SOTOPIA:**

SOTOPIA is a general-domain, open-ended interactive simulation platform for evaluating LLMs' social intelligence. It features:

- 90 real-world-inspired scenarios (negotiation, collaboration, persuasion, etc.)
- 40 richly profiled agents (values, personality, secrets, etc.)
- Social goals with conflicting incentives.



In each episode, SOTOPIA first samples a social scenario context, goals, and characters, and then assigns a social goal and character to each agent involved. Agents (artificial agents or humans) in SOTOPIA role-play characters while attempting to achieve their goals. The agents' performance is evaluated through a multi-dimensional framework, SOTOPIA-EVAL.

SOTOPIA-EVAL assesses models along 7 dimensions: Goal completion, Believability, Knowledge acquisition, Secret-keeping, Relationship effects, Social rule adherence, and Financial/material outcomes.

**Goal Completion (GOAL)** [0–10] is the extent to which the agent achieved their goals. Agents' social goals, defined by the environment, are the primary drivers of their behavior [6]

**Believability (BEL)** [0–10] focuses on the extent to which the agent's behavior is perceived as natural, realistic, and aligned with the agents' character profile, thus simulating believable proxies of human behavior [7]. Specifically, we consider the following criteria: 1. If the agent interacts with others in a natural and realistic manner (naturalness). 2. If the actions of the agent align with their character traits e.g., personality, values, etc. (consistency).

**Knowledge (KNO)** [0–10] captures the agent's ability to actively acquire new information. This dimension is motivated by the fact that curiosity, i.e., the desire to know or learn, is a fundamental human trait [8, 9]. Specifically, we consider the following criteria: What information the agent has gained through the interaction, whether the information the agent has gained is new to them, and whether the information the agent has gained is important to them.

**Secret (SEC)** [-10-0] measures the need for agents (humans) to keep their secretive information or intention private [8]. From a game theory perspective, leaking secrets often leads to a loss of utility [10]. However, revealing secrets can be a powerful tool to build trust and thus improve relationships [11]. In this dimension, we ask what secret or secretive intention the participant wants to keep, and whether they keep it successfully.

**Relationship (REL)** [-5-5] captures the fundamental human need for social connection and belonging [9, 12]. In this dimension, we ask what relationship the participant has with the other agent(s) before the interaction, and then evaluate if the agents' interactions with others help preserve or enhance their personal relationships. Additionally, we ascertain whether these interactions also impact the social status or the reputation of the agent.

**Social Rules (SOC)** [-10-0] concerns norms, regulations, institutional arrangements, and rituals. We differentiate between two types of social rules: social norms and legal rules. Legal rules encompass prohibited actions and the potential for punishment by institutionalized force, while social norms encompass normative social rules (e.g., it is considered rude to speak loudly in a library).

**Financial and Material Benefits (FIN)** [-5-5] pertains to traditional economic utilities as addressed by classic game theory [10, 13]. We consider financial utility to be comprised of both short-term monetary benefits (e.g., earnings) and long-term economic payoffs (e.g., job security, stock holdings, funding opportunities)

They benchmark GPT-4, GPT-3.5, LLaMA-2, and MPT, and compare performance to human participants. They show that models underperform humans in hard tasks, especially in strategic reasoning and efficiency (e.g., GPT-4 uses 3x more words per turn).

Results:

| Dim. | Range    | GPT-4       | GPT-3.5 | Llama-2 | MPT   |
|------|----------|-------------|---------|---------|-------|
| SOC  | [-10, 0] | -0.07       | -0.08   | -0.11   | -0.09 |
| SEC  | [-10, 0] | -0.14       | -0.08   | -0.14   | -0.07 |
| FIN  | [-5, 5]  | <b>0.81</b> | 0.46    | 0.40    | 0.28  |
| REL  | [-5, 5]  | <b>1.94</b> | 1.23    | 0.91    | 0.58  |
| KNO  | [0, 10]  | <b>3.73</b> | 3.40    | 3.11    | 2.11  |
| GOAL | [0, 10]  | <b>7.62</b> | 6.45    | 5.38    | 4.10  |
| BEL  | [0, 10]  | <b>9.28</b> | 9.15    | 8.10    | 6.17  |

Above is aggregated performance of each model by averaging across different partner models. The best performance for each dimension is bolded when significantly better than the second best in t-test ( $p < 0.05$ ).

The results demonstrate that GPT-4 could automate the evaluation of agent performance based on SOTOPIA-EVAL. Building on this, we show that SOTOPIA can be used for understanding not



only the differences among models but also the difference between models and humans in terms of social interaction abilities.

### **Limitations:**

#### **Generative Agents:**

1. Memory Fragility: The architecture depends on heuristic memory retrieval which may fail or hallucinate under distribution shifts.
2. Scalability: Only 25 agents in a toy world; unclear how this generalizes to large, open-world environments.
3. Style Overfit: Language style reflects LLM training data (e.g., formal speech), limiting character realism.
4. Ethical Risks: Emergent parasocial relationships and manipulative behavior raise safety concerns.

#### **SOTOPIA:**

1. Evaluator Bias: GPT-4 is both an agent and evaluator; this may introduce self-reinforcing biases.
2. Limited Multimodality: All actions are text-based; no embodied or visual reasoning tested.
3. Secret and Norm Dimensions: These are negatively scored only, which may oversimplify trust-building strategies.
4. Low Robustness to Partner Models: Weaker conversational partners significantly degrade performance.

### **Summary of Class Discussion:**

P1 Q1 we discuss the *lack of concrete downstream tasks* or benchmarks of the paper. While the simulation of agent behavior is impressive, the implementation of such agents—even in a controlled, manually designed game environment—remains technically difficult. The system's reliance on handcrafted scenarios and hard-coded elements also limits its broader applicability.

This led to a deeper conversation on the *limitations of current LLMs in role-playing diverse characters*. We discussed how methods like retrieval-augmented generation (RAG) or memory-augmented systems are not reliably effective in dynamic or real-time settings, such as games, let alone real-world environments. Furthermore, the ability of LLMs to maintain distinct personalities is currently shallow—often constrained to surface-level language traits. Without extensive fine-tuning or carefully designed control mechanisms, these personalities tend to converge back into generic, default LLM behavior.

We also discussed that the emergent ability mentioned in the paper is somewhat abstract, it is not a practical ability such as lifting a water bucket, but rather like information diffusion, information distribution, etc.

Then the discussion evolved to the incapability of LLM roleplaying different characters. Many methods such as RAG-based system or memory-based system are not as effective in gaming



environments (not to say the real world), and the personality are often constrained within language abilities. And, even the language-personality quickly converges back into vanilla llm behavior without careful finetuning or many hard-coded rules.

P1 Q2, one small question was how the paper assigns distinct personas to each agent. The answer provided is that each agent is initialized with a hard-coded backstory, which is passed as their initial memory. These scripted biographies inform the agent's initial behavior, but again, their influence tends to diminish over time without reinforcement.

P2 Q1 Ask about the “validity” of the evaluation, especially about the direction of LLM-as-judge. Specifically, the discussion focused on whether GPT-4's role as an evaluator introduces systemic biases, especially when it's used to judge its own outputs or those of similar models. This concern extended to a broader critique of the current challenges in model validation and transparency. We discussed that part of the difficulty in releasing the next generation of models (GPT-5) may stem from the limitations of using LLMs themselves as judges and from the diminishing returns of synthetic data. Since existing models like GPT-4 have already been used extensively to bootstrap and generate large volumes of training and evaluation data, it's possible that further performance gains are becoming increasingly difficult to achieve. In other words, the reliance on self-generated data and self-evaluation may have led to a bottleneck in both benchmarking and model improvement.

P2 Q2 we then discussed why do we care about the social aspect of llm, we agreed that social competence is increasingly relevant for AI in real-world applications, especially in domains like education, negotiation, therapy, or customer support. Although current LLMs are primarily tools, they are rapidly evolving into roles like companions, mentors, and collaborators. However, language alone can only encode a limited view of social context, which constrains models' social awareness and ability to adapt to nuanced interpersonal situations.

P2 Q3, one particular discussion was about the word “believability”. Some initially interpreted this as a measure of *trustworthiness* or *moral alignment*, while the paper defines it more narrowly (whether an agent's behavior appears natural and contextually fitting). This sparked a broader conversation on whether we can—or should—try to assess *intentions* in LLMs. Given that many psychological and linguistic definitions of trust or deception rely on the speaker's intent, it's still unclear how we can apply such frameworks to inherently stateless models like LLMs.

## Other Discussion

We also briefly discussed the Arxiv Copilot [14] project by the ULab, which was vaguely *inspired in spirit* by the two papers. It aims to build LLMs that can *role-play specific public figures* (e.g., famous AI researchers), offering engaging and informative conversations. However, some challenge still exists, such as *legal challenges*, privacy concerns, and repetitive names.

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